

WEEK 5 LAB

The neurological exam.

2 to 3 hours. Graded. Investigate It, 25 percent of your grade across the term.

This week you run the parts of a real neurological exam on a partner: mental status, the cranial nerves, motor, reflexes, sensory, and coordination and gait. Then you test the special senses by hand. For every part you write down what the test checks and what normal looks like, and you draw the pathway it tests. Then you watch one patient with an abnormal finding in the University of Utah NeuroLogic Exam videos and work out where the problem is.

The videos come from NeuroLogic Exam: An Anatomical Approach, by the University of Utah, shared under a Creative Commons license. The whole site, listed by exam. Each part of the exam has a Normal Exam and an Abnormal Examples page.

What you need

- A partner who agrees to be examined. A family member or a friend is fine. You are learning the exam, not diagnosing anyone.
- Your phone flashlight, a tissue, coffee or soap to smell, a flavored jelly bean or hard candy, a clean toothpick or an opened paperclip, a capped pen, a ruler, and a rubber kitchen spatula if you have one.
- Paper and a pen for the five drawings, or the printed worksheet. You can type everything else on this page.

This is practice of an exam, not a medical test, and nothing here can tell you whether anything is wrong with your partner. If you notice something that worries you, that is a question for a clinician.

Your name

Your name

Who you examined (first name or relationship is enough)

Part 1. Mental status

About 5 minutes

Every neurological exam starts here, because a person who is confused cannot give reliable answers on the rest. You are checking the cerebral cortex as a whole.

What you test, and what is normal

TEST	WHAT IT CHECKS	NORMAL	YOUR PARTNER
Orientation	Ask their name, where they are, the date, and why they are here	Correct on all four: person, place, time, situation	
Attention	Spell WORLD backward, or count back from 100 by 7s	D L R O W, or 93, 86, 79, 72, 65	
Memory	Say three words, then ask for them again after 5 minutes	All three recalled	
Language	Name two objects you point to, and repeat a sentence	Names both, repeats it exactly	

One abnormal finding, and what it means

Open the NeuroLogic Exam site index, find Mental Status Exam, and choose Abnormal Examples. Pick one patient. This week’s patient, Camila, was charted with a Glasgow Coma Scale of 13 on her first morning, so watch for what a reduced level of consciousness looks like.

What did you see? Describe the finding in plain words, and say which side.

Which structure is damaged, and on which side? Name the nerve, nucleus or tract, and where its cell bodies are.

Is this an upper motor neuron, lower motor neuron, sensory or cerebellar problem? What in the finding tells you?

Part 2. Cranial nerves

About 20 minutes

The twelve cranial nerves carry sensation and movement for the head and neck, and Table 10.1 in your textbook lists each one as sensory, motor or mixed. Most of their nuclei, the clusters of cell bodies in the brainstem, sit in a predictable order from top to bottom: III and IV in the midbrain, V, VI, VII and VIII in the pons, and IX, X and XII in the medulla. CN I and II connect straight to the forebrain, and CN XI comes from the upper cervical spinal cord. Watch the normal cranial nerve exam videos before you try it on your partner.

Use your phone flashlight for the pupils only for a second or two at a time. Skip the gag reflex: you will see it in the video instead.

What you test, and what is normal

NERVE	HOW YOU TEST IT	WHERE THE CELL BODIES ARE	NORMAL	YOUR PARTNER
I olfactory	Eyes closed, one nostril blocked, identify coffee or soap	Olfactory receptor neurons in the nasal epithelium	Identifies it, each side	
II optic	Read small print one eye at a time; count fingers in each quarter of their vision; light in one eye	Retinal ganglion cells	Reads it; sees fingers in all four quarters; both pupils constrict	
III, IV, VI	Follow your finger in an H with their eyes, head still	III and IV: midbrain. VI: pons	Both eyes move together fully, no double vision	
V trigeminal	Light touch on forehead, cheek and jaw; clench the teeth and feel the masseter	Sensory: trigeminal ganglion. Motor: pons	Feels all three areas equally; masseters firm on both sides	
VII facial	Raise the eyebrows, close the eyes tight, smile, puff the cheeks	Motor nucleus in the pons	Both sides move equally	
VIII vestibulocochlear	Rub your fingers beside each ear	Spiral and vestibular ganglia; nuclei at the pons and medulla	Hears it equally on both sides	
IX, X	Say ah and watch the soft palate and uvula	Medulla	Palate rises evenly, uvula stays in the midline	
XI accessory	Shrug against your hands; turn the head against your hand	Upper cervical spinal cord	Strong and equal	
XII hypoglossal	Stick the tongue straight out, then push it into each cheek	Medulla	Tongue comes out in the midline, strong both ways	

Draw the pathway

Draw it on paper: a brainstem from the side as three stacked boxes, midbrain, pons and medulla, with the upper cervical cord below. Write each cranial nerve from III to XII beside the level where its nucleus sits.

One abnormal finding, and what it means

Open the abnormal cranial nerve exam videos and pick one. Good choices: the 6th nerve palsy and the 3rd nerve palsy under Versions, the 12th nerve lesion, or the 9th and 10th nerve deficit. A cranial nerve or its nucleus is a lower motor neuron, so think about which side the weakness is on.

What did you see? Describe the finding in plain words, and say which side.

Which structure is damaged, and on which side? Name the nerve, nucleus or tract, and where its cell bodies are.

Is this an upper motor neuron, lower motor neuron, sensory or cerebellar problem? What in the finding tells you?

Part 3. Motor

About 10 minutes

The motor exam follows the corticospinal pathway from the last unit: an upper motor neuron with its cell body in the motor cortex, crossing at the bottom of the medulla, and a lower motor neuron with its cell body in the ventral horn of the spinal cord. Look first, then feel, then test.

Strength is graded from 0 to 5: 0 no contraction, 1 a flicker, 2 moves only with gravity removed, 3 moves against gravity, 4 moves against some resistance, 5 normal strength.

What you test, and what is normal

TEST	HOW YOU DO IT	NORMAL	YOUR PARTNER
Bulk	Compare the muscles of the two sides by eye	Equal, no wasting, no twitching under the skin	
Tone	Partner relaxed; bend and straighten the elbow and knee slowly	Slight, even resistance; not floppy, not stiff	
Strength	Push against shoulders, elbows, wrists, hips, knees and ankles, one side then the other	5 of 5, equal on both sides	
Pronator drift	Arms straight out, palms up, eyes closed, for 20 seconds	Both arms stay level, palms stay up	

Draw the pathway

Draw it on paper: the corticospinal pathway from the motor cortex to a muscle of the right hand. Mark the upper motor neuron's cell body, where it crosses, and the lower motor neuron's cell body.

One abnormal finding, and what it means

Watch an abnormal example from the site index: under Motor Exam, choose Abnormal Examples. Ask yourself whether the weakness comes with high tone and brisk reflexes, or with low tone, wasting and twitching. That one question separates upper from lower motor neuron.

What did you see? Describe the finding in plain words, and say which side.

Which structure is damaged, and on which side? Name the nerve, nucleus or tract, and where its cell bodies are.

Is this an upper motor neuron, lower motor neuron, sensory or cerebellar problem? What in the finding tells you?

Part 4. Reflexes and reaction time

About 20 minutes

Each tendon reflex tests one reflex arc and the segments of the cord that run it. Tap the tendon briskly with the side of your fingers or the rubber edge of a kitchen spatula. Grade each reflex 0 to 4+: 0 absent, 1+ less than usual, 2+ normal, 3+ brisker than usual, 4+ very brisk, often with clonus. Compare left with right.

If a reflex is hard to get, try the Jendrassik maneuver: your partner hooks the fingers of both hands and pulls hard at the moment you tap.

What you test, and what is normal

REFLEX	WHERE YOU TAP	CORD SEGMENTS	NORMAL	YOUR PARTNER
Biceps	Your thumb on the biceps tendon at the inner elbow; tap your thumb	C5 and C6	2+, elbow bends a little	
Triceps	Just above the elbow at the back, arm hanging bent	C7	2+, elbow straightens a little	
Patellar (knee jerk)	Just below the kneecap, legs dangling	L2 to L4	2+, the leg kicks	
Achilles (ankle jerk)	The heel cord, foot held slightly bent up	S1	2+, the foot points down	
Plantar response	Stroke the outer edge of the sole firmly, heel to toes, with a capped pen	Corticospinal tract and S1	Toes curl down	
Ruler drop	Drop a ruler between their fingers without warning, 5 tries; record the average in cm	Eyes, brain, cord, hand	About 10 to 20 cm, which is about 140 to 200 ms	

Draw the pathway

Draw it on paper: the reflex arc for the knee jerk on a spinal cord cross section, with all five parts labeled and the synapse counted. Beside it, draw the much longer path for catching the ruler.

One abnormal finding, and what it means

Open the abnormal motor exam videos and watch the tendon reflexes or the pathological reflexes. Convert your ruler distance to time with milliseconds = 45.15 times the square root of the centimeters, and in your interpretation say why a reflex can never be as slow as your reaction time.

What did you see? Describe the finding in plain words, and say which side.

Which structure is damaged, and on which side? Name the nerve, nucleus or tract, and where its cell bodies are.

Is this an upper motor neuron, lower motor neuron, sensory or cerebellar problem? What in the finding tells you?

Part 5. Sensory

About 15 minutes

Two pathways carry body sensation, and the exam tests them separately. Pain and temperature travel in the spinothalamic tract, which crosses in the spinal cord. Fine touch, vibration and joint position travel in the dorsal columns, which cross in the medulla. Both start with a sensory neuron whose cell body is in a dorsal root ganglion. Keep your partner's eyes closed for every test.

For sharp, use the point of a clean toothpick or an opened paperclip and press only until it feels sharp. Never break the skin. Use a fresh point for each person.

What you test, and what is normal

TEST	HOW YOU DO IT	PATHWAY	NORMAL	YOUR PARTNER
Light touch	A wisp of tissue on the back of each hand and the top of each foot	Dorsal columns, with some in the spinothalamic tract	Feels it everywhere, equal on both sides	
Sharp or dull	Mix the point and the blunt end; partner says sharp or dull	Spinothalamic	Correct every time, both sides	
Joint position	Hold the big toe by its sides and move it a little up or down	Dorsal columns	Says up or down correctly	
Two-point discrimination	Two points of a bent paperclip at the fingertip and the forearm; find the smallest gap felt as two	Dorsal columns; small receptive fields	A few millimeters at the fingertip, several centimeters on the forearm	
Romberg test	Feet together, arms at the sides, eyes open then closed, for 30 seconds each, standing right next to you	Joint position sense, with vision and balance	Steady with the eyes closed too	

Draw the pathway

Draw it on paper: the spinothalamic tract and the dorsal column pathway from the right foot to the cortex, in two colors. Mark each dorsal root ganglion and the level where each pathway crosses.

One abnormal finding, and what it means

In the site index, under Sensory Exam, choose Abnormal Examples and pick one. A Romberg test that is steady with the eyes open but unsteady with them closed points to lost joint position sense, not to the cerebellum. Decide which pathway your example has lost.

What did you see? Describe the finding in plain words, and say which side.

Which structure is damaged, and on which side? Name the nerve, nucleus or tract, and where its cell bodies are.

Is this an upper motor neuron, lower motor neuron, sensory or cerebellar problem? What in the finding tells you?

Part 6. Coordination and gait

About 10 minutes

The cerebellum smooths and times movement, and each side of the cerebellum controls the same side of the body. These tests look for errors in aim, rhythm and balance. Stand beside your partner during the gait tests.

What you test, and what is normal

TEST	HOW YOU DO IT	NORMAL	YOUR PARTNER
Finger to nose	Touch your finger, then their own nose, back and forth, as you move your finger	Smooth and accurate, no shaking as the finger arrives	
Rapid alternating movements	Pat the thigh with the palm, then the back of the hand, as fast as possible	Fast and even on both sides	
Heel to shin	Lying or sitting, run one heel down the other shin	Stays on the shin, smooth	
Gait	Walk normally, then on the heels, then on the toes	Even steps, arms swing, steady	
Tandem walk	Walk heel to toe along a straight line	Stays on the line without stepping off	

One abnormal finding, and what it means

In the site index, under Coordination Exam or Gait Exam, choose Abnormal Examples and pick one. If the problem is on one side, decide which side of the cerebellum is damaged, and remember that, unlike the motor cortex, each side of the cerebellum works on its own side of the body.

What did you see? Describe the finding in plain words, and say which side.

Which structure is damaged, and on which side? Name the nerve, nucleus or tract, and where its cell bodies are.

Is this an upper motor neuron, lower motor neuron, sensory or cerebellar problem? What in the finding tells you?

Part 7. The special senses

About 25 minutes

PhysioEx has no exercise on the special senses, so this part is hands-on. Each test shows one idea from the vision and hearing, balance, taste and smell walkthroughs. Do each test on yourself first, then on your partner.

Shine the flashlight for a second or two at a time. Do the balance test standing next to a wall or a counter you can grab.



What you test, and what is normal

TEST	HOW YOU DO IT	WHAT IT SHOWS	NORMAL	YOUR PARTNER
Blind spot	Cover the left eye. Stare at the + above with the right eye, at arm's length, and slowly bring the page closer. Measure the distance when the dot disappears	The optic disc, where the optic nerve leaves, has no photoreceptors	The dot vanishes at one distance, then comes back as you get closer	
Near point	Cover one eye. Bring small print toward the open eye until it blurs. Measure eye to page in cm	Accommodation: how round the lens can get	About 10 cm at age 20, farther with age as the lens stiffens	
Pupils, light	Dim room. Shine the light into one eye and watch both pupils	Pupillary light reflex: CN II in, midbrain, CN III out to both eyes	Both pupils constrict, equally	
Pupils, near	Have your partner look from across the room to your fingertip 20 cm from their nose	The near response: eyes turn in, pupils constrict, lens rounds up	Both eyes turn in and both pupils get smaller	
Humming Weber	Hum steadily. Then press a finger into one ear canal and keep humming	Blocking the canal imitates a conductive hearing loss	Centered with both ears open; louder in the blocked ear	
Flavor with the nose held	Hold the nose, chew a flavored jelly bean or candy, and name the flavor. Then let go	Flavor needs smell as well as taste	Only sweet with the nose held; the flavor appears when you let go	
Smell adaptation	Sniff coffee or soap every 10 seconds for 1 minute; rate it 0 to 10 each time	Olfactory receptors adapt to a steady odor	The rating falls over the minute	
One-leg stand	Stand on one leg beside a wall, eyes open, then eyes closed; time each up to 30 seconds	Balance combines vision, the vestibular system and joint position sense	Shorter with the eyes closed	

Draw the pathway

Draw it on paper: the visual pathway from both eyes through the chiasm to the cortex, and mark where the blind spot is in the right eye's retina and in its visual field.

One abnormal finding, and what it means

Open the abnormal cranial nerve exam videos and watch the Cranial Nerve 2 visual fields example or the Cranial Nerve 8 Weber and Rinne example. Use what your own humming Weber test showed you to explain which way the sound goes and what that says about the lesion.

What did you see? Describe the finding in plain words, and say which side.

Which structure is damaged, and on which side? Name the nerve, nucleus or tract, and where its cell bodies are.

Is this an upper motor neuron, lower motor neuron, sensory or cerebellar problem? What in the finding tells you?

Look back at your thinking

Before you started, which part of the exam did you expect to be hardest to do, and why?

Which abnormal finding was hardest to place, and what finally told you where the problem was?

Name one thing you would do differently the next time you examine someone.

Turn it in

Save this page as a PDF with the button below. Photograph your five drawings. If you worked on the printed worksheet, scan or photograph every page instead. Combine the PDF and the photos into one PDF and upload it to the Week 5 lab assignment in Canvas. Due Sunday, October 11 at 10:00 pm.